

Theory of Computation

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Part = 4

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Subject = Theory of Computation

Semester = 4th

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Unit = 3 (Regular Expression)

Important Questions:

1. Kleene Closure Operator
2. Positive Closure

$$\text{if } \Sigma = \{a, b\}$$

• Kleene Closure

$$\Sigma^* = \{a, b\}^* = \{\epsilon, a, b, ab, aa, bb, ba, \dots\}$$

Positive Closure

$$\Sigma^+ = \{a, b\}^+ = \{a, b, ab, bb, aa, ba, \dots\}$$

Given $\Sigma = \{a, b\}$ write the regular expression for the following language that contains

- a) All the strings have a single 'b'

Soln,

$$a^* b a^*$$

- b) All string have at least one 'b'

Soln

$$(a+b)^* b (a+b)^*$$

note:

$$(a+b) = a \cup b$$

$$(a, b) = a \cup b$$

$$\boxed{\epsilon b \epsilon \rightarrow b}$$

c) All string having bbbb as substring

Soln,

$$(a+b)^* bbbb (a+b)^*$$

d) All string ending with ab

Soln

$$(a+b)^* ab$$

e) All string starting with ba

Soln

$$ba (a+b)^*$$

f) All string beginning & ending with a.

Soln,

$$a (a+b)^* a$$

g) All string containing 2 consecutive 'b'

$$(a+b)^* bb (a+b)^*$$

Given the regular expression for following over alphabet (0,1)

a) Set of all string with 2nd symbol from right is 1

$$(0+1)^* 1 (0+1)^*$$

$$\dots \frac{1}{2} \dots$$

b) Set of all strings starting with 00 or 11 & ending with 10 or 01
 soln,

$$00 \text{ --- } 11 \\ (00 + 11) (0 + 1)^* (10 + 01)$$

Given the regular expression for the following language over alphabet $\{a, b\}$

a) Set of strings all strings with substring bab or abb

$$(a+b)^* (bab + abb) (a+b)^*$$

or,

$$(a+b)^* bab (a+b)^* + (a+b)^* abb (a+b)^*$$

b) Set of all strings whose 3rd symbol is 'a' and 5th symbol is b
 soln,

$$(a+b) (a+b) a (a+b) b (a+b)^*$$

conversion of Regular Expression to finite Automata

R.E $\rightarrow$ F.A
 i.e R.E $\rightarrow$ ϵ -NFA

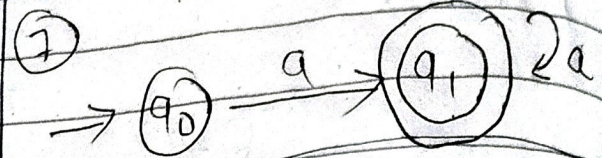
Regular Expression	Finite Automata
1. ϕ	① $\rightarrow$ (q ₀)
2. ϵ	② $\rightarrow$ (q ₀)
3. a	③ $\rightarrow$ (q ₀) $\xrightarrow{a}$ (q ₁)
4. b	④ $\rightarrow$ (q ₁) $\xrightarrow{b}$ (q ₁)
5. a + b	⑤ $\rightarrow$ (q ₀) $\xrightarrow{a}$ (q ₁) $\xrightarrow{b}$ (q ₁) Or, $\rightarrow$ (q ₀) $\xrightarrow{a, b}$ (q ₁)
⑥ a* i.e $\{ \epsilon, a, aa, aaa, \dots \}$	⑥ $\rightarrow$ (q ₀) $\xrightarrow{a}$ (q ₀)

Regular expression

Finite automata

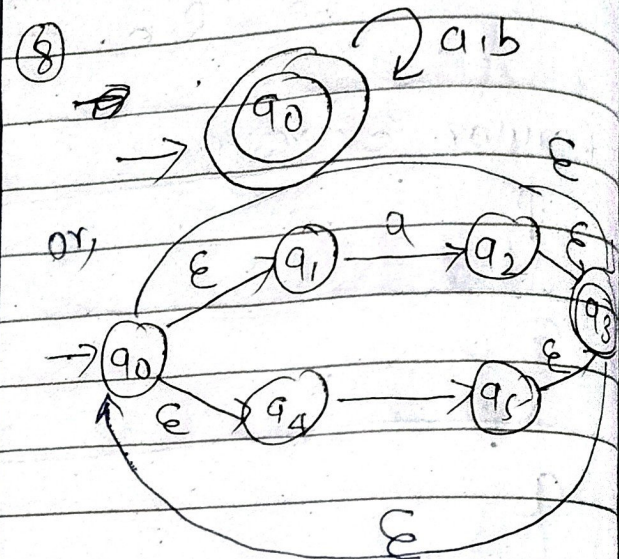
7 a^+

i.e. $\{a, aa, aaa, \dots\}$



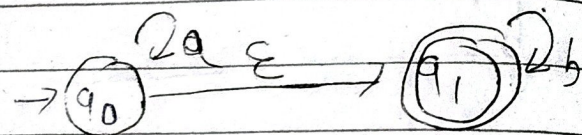
8 $(a+b)^*$

i.e. $\{\epsilon, a, b, aaa, bbb, abab, \dots\}$



9 a^*b^*

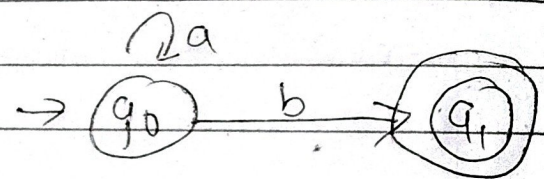
i.e. $\{\epsilon, a, b, ab, aaabbb, aabbbb, \dots\}$



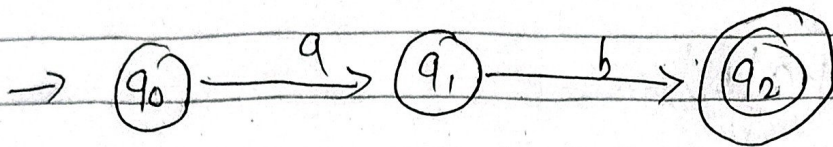
10 a^*b

i.e. $\{\epsilon, a, b, aa, \dots\}$

i.e. $\{b, ab, aab, aaab, \dots\}$

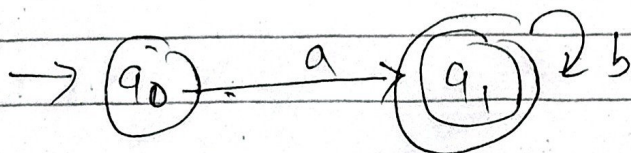


11 ab



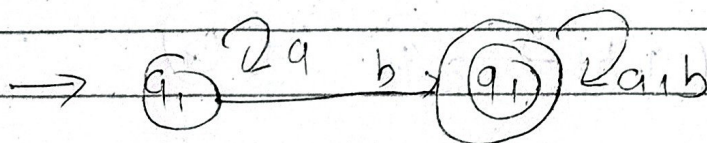
12 ab^*

ie $\{a, ab, abb, abbb, \dots\}$



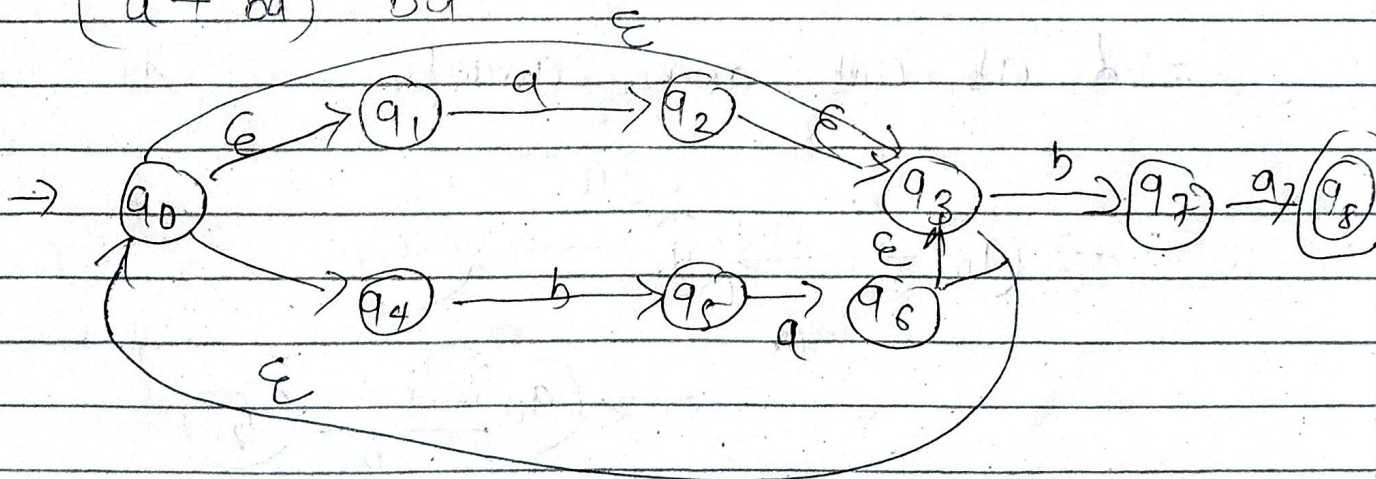
13

$a^*b(a+b)^*$



14

$(a+ba)^*ba$



Convert the following Regular Expression to its equivalent finite Automata.

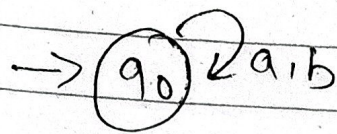
$$(a+b)^* (abb + a^+b)$$

Soln,

for making easier, let's divide given expression in these parts as

$(a+b)^*$, abb , a^+b and draw for each separately.

$$(a+b)^*$$

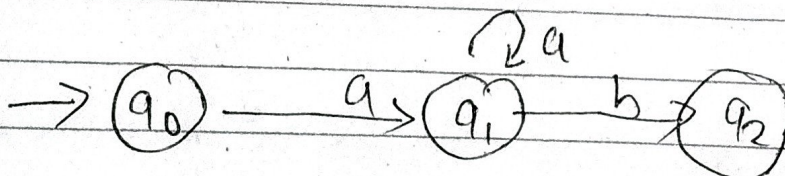


$$abb$$

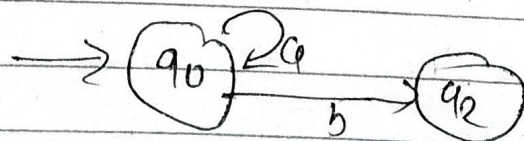


$$a^+b$$

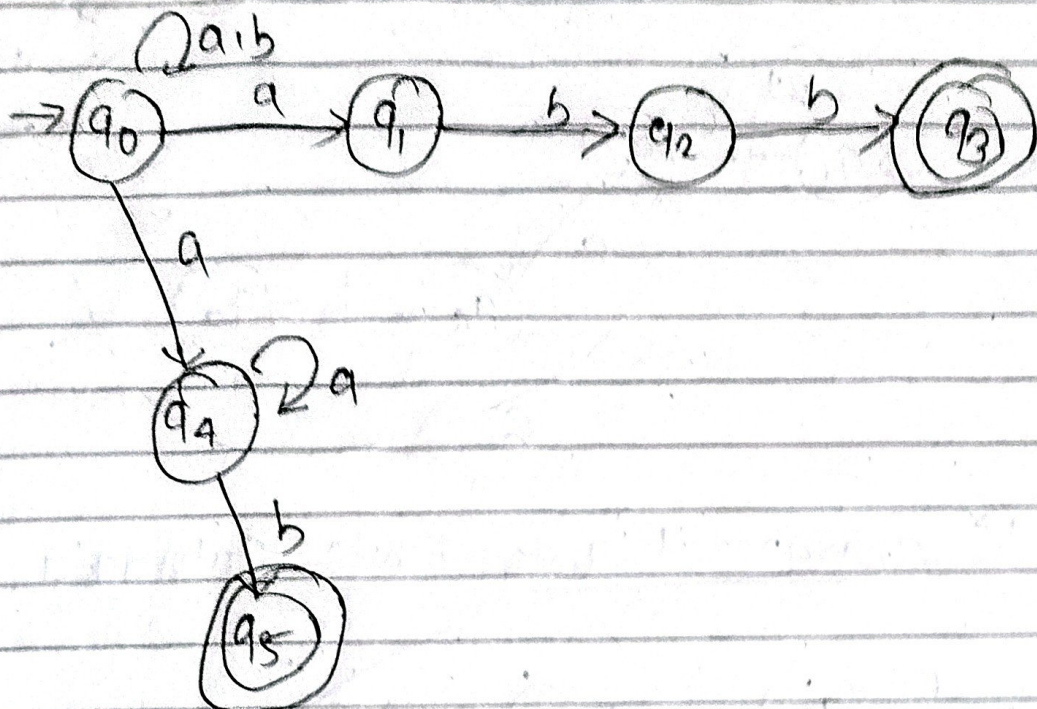
$$= \{ ab, aab, aaab, a^4ab, \dots \}$$



or,



Now we combine all of them in one as follows

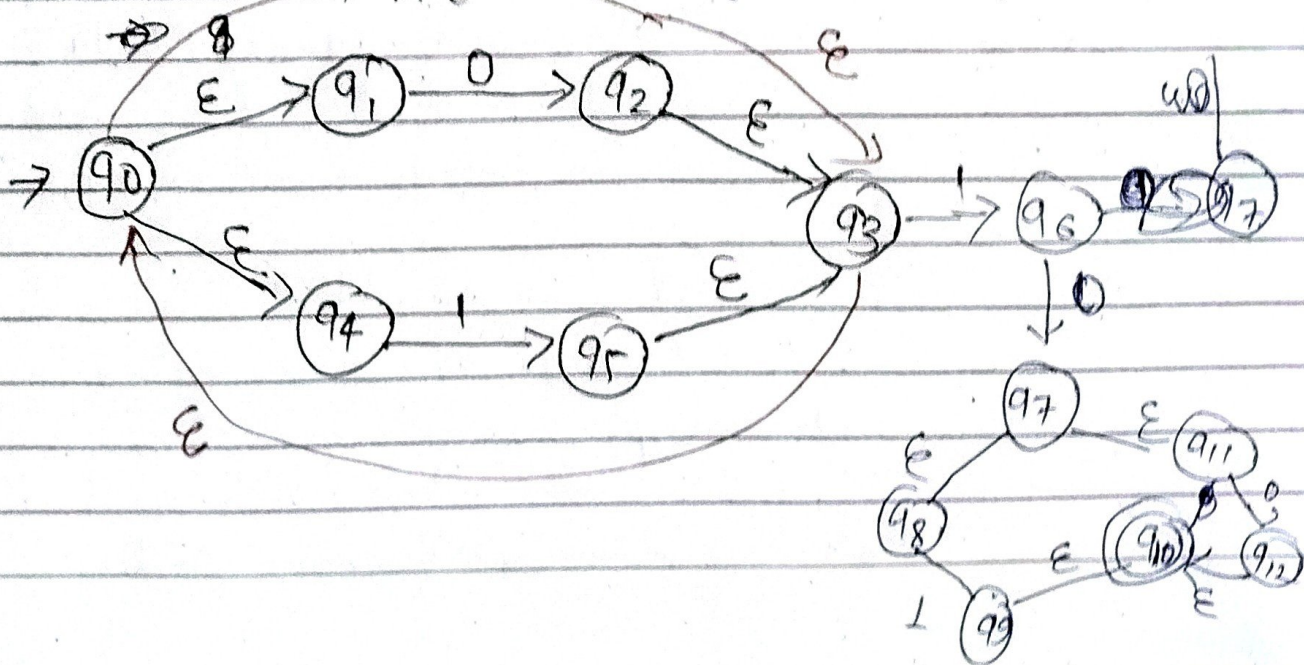


convert the following regular expression into equivalent finite Automata:

① $(0+1)^* 10 (1+0)$

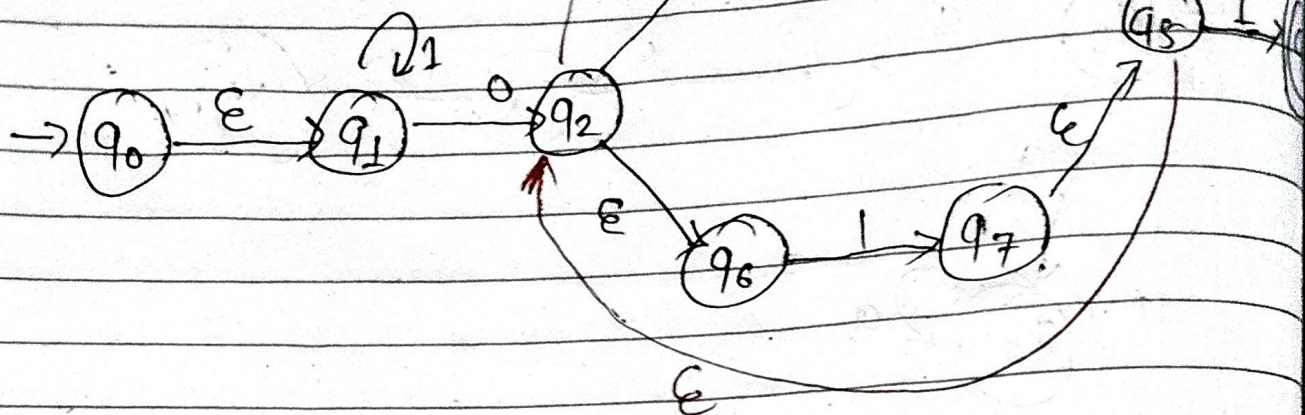
Soln,

$\hookrightarrow$ R.E to F.A (ie E-NFA)

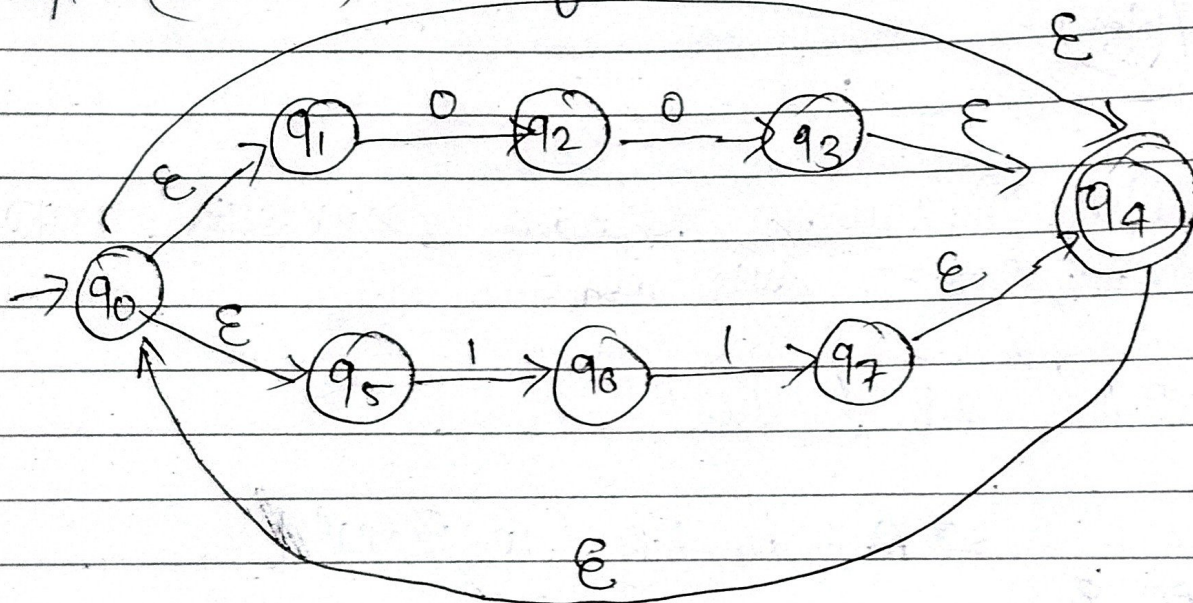


⑤ $1^*0(0+1)^*1$

soln



$(00+11)^*$ change it into finite Automata.



Ardens Theorem:

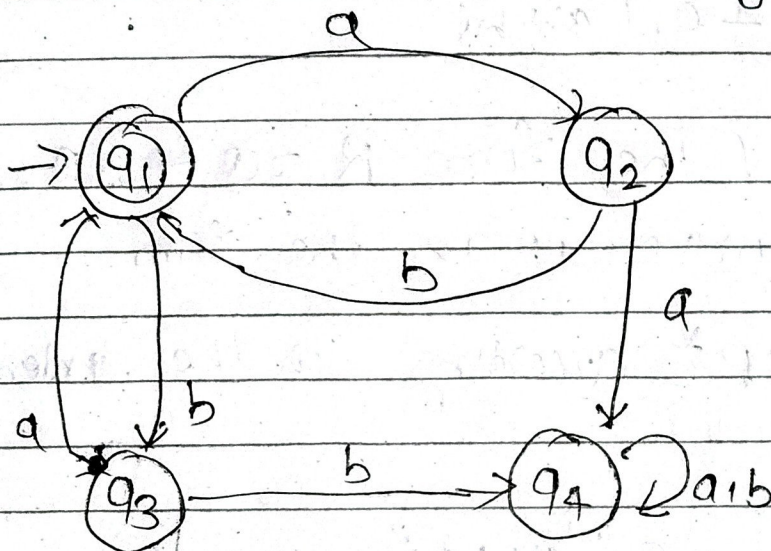
Equation;

$$R = \emptyset + RP$$

i.e $R = \emptyset P^*$

eg = 1

Conversion of DFA to Regular Expression



Soln

here, equations for each state based on incoming transition are as follow:

$$q_1 = \epsilon + q_2 b + q_3 a \quad \text{--- (i)}$$

$$q_2 = q_1 a \quad \text{--- (ii)}$$

$$q_3 = q_1 b \quad \text{--- (iii)}$$

$$q_4 = q_2 a + q_3 b + q_4 a + q_4 b \quad \text{--- (iv)}$$

Now, we take eqn (i) (ie final state q_1)

$$q_1 = \epsilon + q_2 b + q_3 a$$

Putting the value of q_2 & q_3 from eqn (ii) & eqn (iii)

$$q_1 = \epsilon + q_1 ab + q_1 bac$$

$$q_1 = \epsilon + q_1 (ab + ba)$$

Now, this is of the form $R = \epsilon + RP$ so, we can write it in the form

$R = RP^*$ according to the ardens theorem,

$$q_1 = \epsilon \cdot (ab + ba)^*$$

$$q_1 = (ab + ba)^*$$

Now, according to the algebraic law for regular expression we know that $\epsilon \cdot R = R$ so, above expr can be written as

$$q_1 = (ab + ba)^*$$

Since q_1 is final state & we get expr for q_1 . Hence $q_1 = (ab + ba)^*$ is the reg. regular expr for given DFA.

Pumping Lemma for regular Expression:

conditions:

- i) $x y z \in A$ For every $i \geq 0$
- ii) $|y| > 0$ (ie length of y should be greater)
- iii) $|xy| \leq p$ (ie length of x & y together should be less than or equal to pumping length)

E.g.

$$x y z \in A$$

Using pumping Lemma Prove that the language

$$A = \{ a^n b^n \mid n \geq 0 \} \text{ is not Regular.}$$

Soln

Assume that A is regular

$$\text{pumping length} = p \text{ (ie } n \text{)}$$

Now,

$$A = a^p b^p$$

Now, we need to divide A into three parts x, y, z

for that let we assume pumping length

$$p = 7$$

then we can write the string A as

$$A = a^7 b^7 = aaaaaaa bbbbbbb$$

Now,

Let us see all the possible ways in which we can divide this A into three parts, x, y, z for that let we take cases as follow

Case = 1 : The y is in the 'a' parts

aaaaaaa bbbbbb
└─┬─┬─┘
x y z

i.e $x = 2a$, $y = 4a$, $z = \text{Remain}$

Case = II. The y is in the 'b' parts

aaaaaaa bbbbbb
└─┬─┬─┘
x y z

i.e

$y = 4b$, $z = 1b$, $x = \text{Remain}$

Now,

For each cases we check condition xy^2z . Let we take $i = 2$,

$$xy^2z$$

Case = I

aa aaaa aaag abbbbbb
└─┘ └─┘ └─┘ └─┘
x y y z

no. of a's = 11

no. of b's = 7

∴ no. of a's $\neq$ no. of b's

So,

$xy^2z \notin A$

Case = II :

aaaaaaaaabbb bbbb bbbb b
└─┘ └─┘ └─┘ └─┘
x y y z

no. of a's = 7

no. of b's = 11

∴ no. of a's $\neq$ no. of b's so $xy^2z \notin A$.

Since this string does not follow the pattern $a^n b^n$ as given by question so, this does not lie in our language. So, we prove that

xy^2z on taking $i=2$ all cases does not lie in our language.

This means A cannot be pumped which leads to Contradiction hence, the language is not regular.

note:

(i) for to prove Regular $i=1$

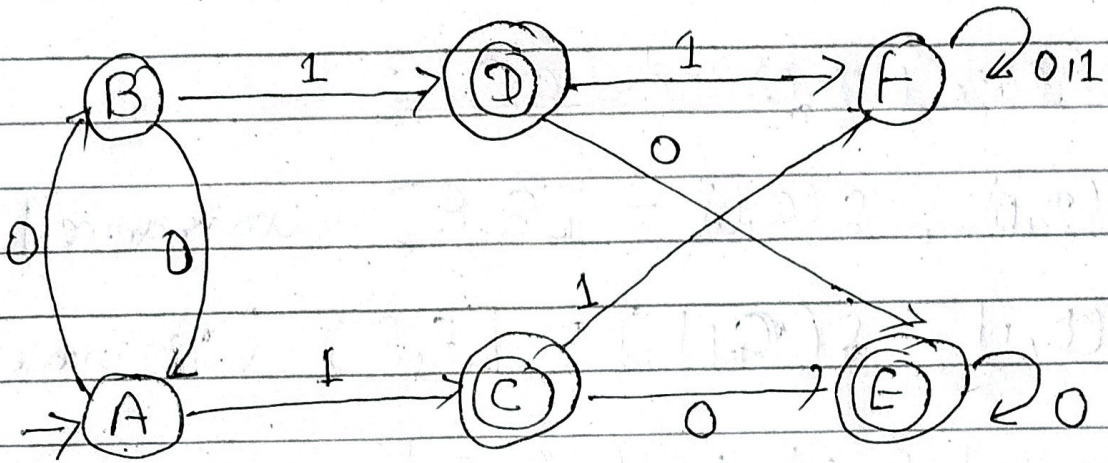
(ii) for to prove not regular $i=2, 3, \dots$

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DFA minimization (Table Filling Algorithm) (myhill - naxode theorem)

eg

minimize the following DFA using Table Filling Algorithm.



Soln

	A	B	C	D	E	F
A						
B						
C	✓	✓				
D	AD	BD	CD			
E	AE	BE	CE	DE		
F	AF	BF	CF	DF		✓

Now, we check the unmarked pair as follow;

For pair (B,A) or (A,B) -

$$[S(B,0), S(A,0)] = [A,B] \text{ (unmarked)}$$

$$[S(B,1), S(A,1)] = [D,C] \text{ (unmarked)}$$

For pair (D,C) or (C,D) :

$$[S(D,0), S(C,0)] = [E,E] \text{ (unmarked)}$$

$$[S(D,1), S(C,1)] = [F,f] \text{ (unmarked)}$$

For pair (E,C) or (C,E)

$$[S(E,0), S(C,0)] = [E,E] \text{ (unmarked)}$$

$$[S(E,1), S(C,1)] = [F,f] \text{ (unmarked)}$$

For pair (D,E) or (E,D)

$$[S(E,0), S(D,0)] = [E,E] \text{ unmarked}$$

$$[S(E,1), S(D,1)] = [F,f] \text{ unmarked}$$

For pair (F, A) or (A, F)

So, the pair (F, A) is distinct
re (F, A) makes
now.

$$[S(F, 0), S(A, 0)] = [F, B] \text{ (unmarked)}$$

$$[S(F, 1), S(A, 1)] = [F, C] \text{ (marked)}$$

For pair (B, F) or (F, B) :

$$[S(F, 0), S(B, 0)] = [F, A] \text{ (marked)}$$

$$[S(F, 1), S(B, 1)] = [F, D] \text{ (marked)}$$

Now, finally we combined unmarked pair (B, A) ; (D, C) , (E, C) , (E, D) as a

single state i.e. (B, A) is now single

state. But in (D, C) , (E, C) , (E, D)

there is common C in (D, C) , (E, C) &

also common E in (E, C) , (E, D) this

show all three 3 pair can also
be combine as single state. which

is illustrated.

